



INDIAN ENGINEERING COMPANY

Authorized Service Center: Crompton Greaves



Marathon, TECO, Baldor, Reliance USA



WEG, HYOSUNG, ABB, TOSHIBA (TMEIC) SHANGHAI, ANSALDO

Rewinding, Refurbishing of AC/DC, LT/HT, 3.3/6.6/11 kV Motors / Alternators / Generators.

613, GIDC ESTATE, RANOLI, P.O. RANOLI, VADODARA – 391 350, GUJARAT, INDIA. Ph: 8401451118, 8401431118.
165/1, OPP. ISHAN EQUIPMENT, RAIKA, N.H. 8, VADODARA – 391340, GUJARAT, INDIA. Ph: 9737072111, 973707911.

E-mail: anil@iecindia.co.in Website: www.iecindia.co.in

ROOT CAUSE ANALYSIS

Date: 08.11.2025

Motor data:

ANSALDO make, 1230 KW, 6.6 KV, 2984 RPM, sr. no. 10208534, Frame size- ETV 630, induction motor

➤ **PROBLEM CHARACTERISTIC:**

Inspection of motor revealed the following problem characteristics:

- Inner bearing cover found damage and rubbing observed on rotor shaft.

➤ **Static Test Result Analysis:**

- Stator winding resistance & inductance checked found balance.
- Stator winding IR value found ok.
- Surge comparison test found normal.



Moving Ahead Together

Management System





INDIAN ENGINEERING COMPANY

Authorized Service Center: Crompton Greaves



Marathon, TECO, Baldor, Reliance USA



WEG, HYOSUNG, ABB, TOSHIBA (TMEIC) SHANGHAI, ANSALDO

Rewinding, Refurbishing of AC/DC, LT/HT, 3.3/6.6/11 kV Motors / Alternators / Generators.

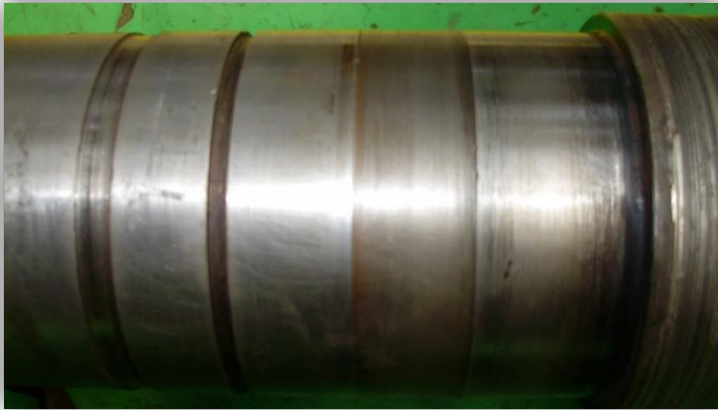
613, GIDC ESTATE, RANOLI, P.O. RANOLI, VADODARA – 391 350, GUJARAT, INDIA. Ph: 8401451118, 8401431118.
165/1, OPP. ISHAN EQUIPMENT, RAIKA, N.H. 8, VADODARA – 391340, GUJARAT, INDIA. Ph: 9737072111, 9737079111.

E-mail: anil@iecindia.co.in

Website: www.iecindia.co.in

➤ **Observation:**

After disassembly the motor, found below observations:



- Punching mark observed in DE side bearing housing roller bearing area.
- On NDE side, end shield bush found loose in bearing housing Insulating ring.

- Dent mark observed on coupling seat.
- DE side bearing found loose on shaft seat and found groove marked on shaft seat.
- DE side IBC seat found rubbed and damaged.



- NDE side bearing found loose on shaft seat.
- NDE side external fan found loose on shaft seat.

Moving Ahead Together

Management System





INDIAN ENGINEERING COMPANY

Authorized Service Center: Crompton Greaves



Marathon, TECO, Baldor, Reliance USA



WEG, HYOSUNG, ABB, TOSHIBA (TMEIC) SHANGHAI, ANSALDO

Rewinding, Refurbishing of AC /DC, LT /HT, 3.3 /6.6 /11 kV Motors / Alternators / Generators.

613, GIDC ESTATE, RANOLI, P.O. RANOLI, VADODARA – 391 350, GUJARAT, INDIA. Ph: 8401451118, 8401431118.
165/1, OPP. ISHAN EQUIPMENT, RAIKA, N.H. 8, VADODARA – 391340, GUJARAT, INDIA. Ph: 9737072111, 973707911.

E-mail: anil@iecindia.co.in

Website: www.iecindia.co.in

- DE side IBC found rubbed, cracked and damage.
- Dent mark observed on DE side labyrinth.



• **History of motor:**

- The said motor was last overhauled in the year of 2023 at M/s. IOCL works.
- The replacement of bearings on both DE and NDE side was carried out. The dimensions were measured of bearing seat on shaft ball bearing 90.03 mm dia. and roller bearing seat 90.00 mm dia.
- The motor was tripped due to found higher DE side bearing temperature on the date of 25.10.2025. And also the vibrations reading of motor was found higher. The vibration reading taken on the date of 15.10.2025.
- M/s. IOCL has shared the history of all activities to IEC.
- The motor was received at IEC workshop at 31.10.2025. and found above observation of the motor.

➤ **Cause of Failure:**

- The main cause of failure of rubbing of IBC on rotor shaft is due to shaft bearing seat worn out. DE and NDE bearing housing which lead to looseness in totality and reduce clearance between IBC inner dia. And shaft IBC seat area.
- Poor workmanship observed during previous job of motor overhauling.
- The IBC was found cracked and damaged the shaft seat due to overheat of the motor.
- The IBC was found rubbed on rotor shaft seat because of improper clearance of bearing seat on shaft.
- Misalignment and looseness also lead to increase of vibration of motor.
- The bearing found loose on shaft seat can cause generate the excessive heat and stress.

Moving Ahead Together

Management System





INDIAN ENGINEERING COMPANY

Authorized Service Center: Crompton Greaves



Marathon, TECO, Baldor, Reliance USA



WEG, HYOSUNG, ABB, TOSHIBA (TMEIC) SHANGHAI, ANSALDO

Rewinding, Refurbishing of AC /DC, LT /HT, 3.3 /6.6 /11 kV Motors / Alternators / Generators.

613, GIDC ESTATE, RANOLI, P.O. RANOLI, VADODARA – 391 350, GUJARAT, INDIA. Ph: 8401451118, 8401431118.
165/1, OPP. ISHAN EQUIPMENT, RAIKA, N.H. 8, VADODARA – 391340, GUJARAT, INDIA. Ph: 9737072111, 973707911.

E-mail: anil@iecindia.co.in Website: www.iecindia.co.in

➤ **Conclusion:**

- Required to be completely overhauling of motor.
- All damaged parts to be replace by new.
- Proper measurement to be carried out of motor and maintained the proper tolerance and clearances of the shaft and end shield housing cover.

➤ **Recommendation:**

- It is recommended that, after 4000 hrs. of running of the motor, the motor shall be overhauled to prevent premature failure.

Moving Ahead Together

Management System



INDIAN ENGINEERING COMPANY



Authorized Service Center: Crompton Greaves



Marathon, TECO, Baldor, Reliance USA



WEG, HYOSUNG, ABB, TOSHIBA (TMEIC) SHANGHAI, ANSALDO

Rewinding, Refurbishing of AC /DC, LT /HT, 3.3 /6.6 /11 kV Motors / Alternators / Generators.

613, GIDC ESTATE, RANOLI, P.O. RANOLI, VADODARA- 391 350, GUJARAT, INDIA. Ph: 8401451118, 8401431118.
165/1, OPP. ISHAN EQUIPMENT, RAIKA, N.H. 8, VADODARA - 391340, GUJARAT, INDIA. Ph: 9737072111, 973707911.

E-mail: anil@iecindia.co.in Website: www.iecindia.co.in

Ref: JOB NO: 25002903

DATE: 02.02.2026

JOINT-INSPECTION

CUSTOMER NAME: I.O.C.L. PRRPL, JATNI, ODISHA

Motor data:-

KW :- 1230	RPM	MAKE :- ANSALDO	G.P. No :- 552
V :- 6600	2984	S.NO :- 102085533	G.P. dtd. :- 28.01.2026
No. of stator slot	54	No. of rotor bar	40
Amp.	124.1	No. of RTD	12 (SIMPLEX)

No.	Item disc.	Observations
1.0	Stator winding	- Stator winding resistance & inductance checked found balance. IR value found ok. - Surge comparison test found ok.
2.0	Rotor	Rotor bar checked found ok (Growler).
3.0	Bearing housing	DE and NDE bearing housing measurement checked - DE side bearing found loose in housing, sleeving to be required. - NDE side bearing found loose in housing and housing found insulated. New insulating housing bush to be replaced by new.
4.0	Rotor shaft	Rotor shaft dimensions checked - DE side coupling found loose on seat. - DE side bearing found loose on seat. - NDE side bearing found loose on seat. - NDE side external fan found loose on seat. Metalizing and machining to be required on all the above mentioned seats on shaft.
5.0	Run out	Checked found ok.
6.0	Dynamic balancing	Rotor dynamic balancing to be carried out.
7.0	Overhauling	Motor overhauling to be carried out.
8.0	Bearings	- New bearings received & supply by M/s IEC: - NU 218 E-XL-M1-C3 (FAG-INDIA) 6218 M/C3 (FAG-GERMANY) - NU 218 E-XL-M1-F1 J 20AAC3 (FAG-ROMANIA) SERVO GEM ELMO PG2 (ES) grease is to be used. Grease supply by M/s IEC.
9.0	Parts	Parts condition checked found - Pulley & key both received. - Shaft lock received.



Moving Ahead Together

02/02/26
SOURABH
CHANDARY





INDIAN ENGINEERING COMPANY

Authorized Service Center: Crompton Greaves



Marathon, TECO, Baldor, Reliance USA



WEG, HYOSUNG, ABB, TOSHIBA (TMEIC) SHANGHAI, ANSALDO

Rewinding, Refurbishing of AC /DC, LT /HT, 3.3 /6.6 /11 kV Motors / Alternators / Generators.

613, GIDC ESTATE, RANOLI, P.O. RANOLI, VADODARA- 391 350, GUJARAT, INDIA. Ph: 8401451118, 8401431118.
165/1, OPP. ISHAN EQUIPMENT, RAIKA, N.H. 8, VADODARA - 391340, GUJARAT, INDIA. Ph: 9737072111, 973707911.

E-mail: anil@iecindia.co.in Website: www.iecindia.co.in

		• Both sides grease pipe received. • Both sides grease collector received. • Both sides internal fan received. • Both sides winding guard received. • Both sides IBC, OBC & LABYRINTH received. • Both the sides V seal required to be replaced.
10.0	Terminal box	• MB received with 1 nos. of cover plate and 3 nos. of insulators received. • SB received with 1 nos. of cover plate and 1 nos. of star link received. • RTD, BTD box received with cover plate. • Space heater box received with cover plate. • CTs terminal box received with cover plate. • From CTs to T.B. connection cable found overheated, required to be replaced.
11.0	BTD	DE side BTD checked 1 pair ok and 1 pair found high value resistance, equivalent BTD required to be replaced. NDE BTD checked found ok (duplex).
12.0	RTD	RTD resistance checked found ok total 12 nos. simplex.
13.0	Space Heater	Space heater resistance checked found ok.
14.0	Fan cover	Received with grill & grill found ok.
15.0	Ext. cooling fan	Received with fan key.

Rep. of IEC

Authorised

Rep. of I.O.C.L. PRRPL, JATNI, ODISHA

Mr. Sourabh Choudhary

[Signature]
02/02/2026

Moving Ahead Together

